YANNIAN NIU

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RESEARCH INTERESTS

My research spans data management, data structures and algorithms, AI for networks, software-defined networking, and AI infrastructure. A growing part of it concerns large language models (LLMs): retrieval infrastructure for LLM serving, and benchmarks that measure how well LLM agents design and optimize systems code. I expect to graduate in May 2027 and am on the job market for full-time positions in these areas.

EDUCATION

University of Connecticut August 2023–Expected May 2027
Ph.D. in Computer Science and Engineering (GPA: 4.00)
Advisor: Dr. Minmei Wang

East China Normal University September 2020–May 2023
Master of Science in Software Engineering

East China Normal University September 2016–May 2020
Bachelor of Engineering in Software Engineering

PUBLICATIONS

(VLDB 2026) Niu Y, Han S, Wang M. STEM^2: A Fast and Space-efficient Data Structure for Exact Multi-Set Membership Queries. 52nd International Conference on Very Large Data Bases.

(SIGMETRICS 2027) Gan Z, Liao S, **Niu Y**, Zheng X, Yu T, Wang M. P4Bench: A Scalable and Adaptive Oracle-Backed Benchmark for P4 Programming with Large Language Models. ACM SIGMETRICS International Conference on Measurement and Modeling of Computer Systems.

(ICNP poster 2026) Liao S, Gan Z, **Niu Y**, Zheng X, Yu T, Wang M. A Property Knowledge Base for Intent-Driven Validation of Programmable Data Planes. The 34th IEEE International Conference on Network Protocols.

(NFV-SDN 2025) Niu Y, Wang M. MagicBox: Towards Consolidated Algorithms of Co-existing Network Functions on Heterogeneous Devices. IEEE Conference on Network Function Virtualization and Software-Defined Networking.

(ToN 2026) Niu Y, Wang M. MagicBox: Towards Consolidated Algorithms of Co-existing Network Functions on Heterogeneous Devices. IEEE Transactions on Networking (extended version). **(Under review)**

(SMC 2023) Zhong J, **Niu Y**, Zhu M. QueryEdge: Real-Time Multi-Video Query in Edge-Cloud Collaborative System. IEEE International Conference on Systems, Man, and Cybernetics.

(SMC 2022) Niu Y, Zhu M. Throughput-Efficient Communication Device Driver for IoT Gateways. IEEE International Conference on Systems, Man, and Cybernetics.

(APNOMS 2022) Niu Y, Zhu M. Assembler: A Throughput-Efficient Module for Network Driver of Edge Computing Gateways. 23rd Asia-Pacific Network Operations and Management Symposium.

RESEARCH EXPERIENCE

Next-Generation Structure for Exact Multi-Set Membership Queries October 2026–Present
Designing a successor to my VLDB 2026 structure for exact multi-set membership queries, targeting higher lookup throughput and substantially cheaper updates. *(Project lead; ongoing.)*

Benchmarking LLM Agents on Constrained Data Structure Design October 2026–Present
Building a benchmark that asks whether LLM agents can produce compact data structures meeting an explicit memory budget, accuracy target, and throughput floor simultaneously, with answers judged by execution against published designs rather than scored by text similarity. *(Project lead; ongoing.)*

Benchmarking LLM Agents on Performance Engineering for Research Code October 2026–Present
Building a benchmark that asks LLM agents to speed up research implementations of compact data structures while requiring that what the code computes stays unchanged, so that correctness and faithfulness to the original design are both checked automatically rather than by human review. *(Project lead; ongoing.)*

Distributed KV Cache Retrieval for LLM Serving July 2026–Present
Adapting our proposed exact multi-set multi-membership indexing solution to enable an LLM serving system to efficiently identify which distributed nodes hold the cached key-value blocks reusable by each request. Because this lookup lies on the critical path of every request, false hits from an approximate filter are particularly costly, as each triggers an unnecessary remote fetch that returns no reusable data. *(Project lead; ongoing.)*

Line-Rate Exact Membership Queries on Programmable Switches July 2026–Present
Developing a P4 implementation of our proposed exact multi-set membership solution for programmable switches, enabling per-packet flow classification and multi-tenant ACL enforcement at Tbps line rates beyond what CPU-based solutions can sustain. The control/data-plane co-design on Intel Tofino addresses pipeline stage budgeting, stateful ALU register packing, and atomic control-plane updates for zero-downtime rule changes. *(Project lead; ongoing.)*

Exact Multi-Set Multi-Membership Query Solution for Metadata Indexing January 2026–Present
Designed an exact multi-set multi-membership index that returns all containers containing a queried key, supporting point queries in log-structured key-value stores and policy checks in network access control. The index avoids both materializing each key's full membership list and probing every set, while providing exact answers with a memory footprint comparable to that of an approximate filter. *(Project lead; ongoing.)*

Intent-Driven Semantic Validation of P4 Programs March 2026–Present
Designed an auditable, LLM-assisted property knowledge base that identifies requirements applicable to a P4 program from authoritative sources and turns them into executable tests for detecting semantic bugs. *(Contributor; ongoing, led by a collaborator in the group.)*

Benchmarking LLM-Based P4 Programming September 2025–July 2026
Built a scalable, adaptive benchmark for evaluating LLM-generated P4 programs, with network-function tasks derived from authoritative sources and packet-level oracles that verify both data-plane code and control-plane configurations on software and hardware switches. *(Contributor; led by a collaborator in the group. Published at SIGMETRICS 2027.)*

Fast and Space-Efficient Exact Multi-Set Membership Queries February 2025–December 2025
Designed an exact multi-set membership query algorithm that outperforms prior works in query throughput, memory cost, and accuracy. Demonstrated its broad applicability as a practical primitive for traffic management and routing, distributed storage and caching, and privacy-preserving systems. *(Published at VLDB 2026.)*

Consolidating Co-existing NFs on Heterogeneous Devices September 2023–January 2025
Designed an end-to-end system that automatically selects network function (NF) algorithms, generates optimized consolidated algorithms, and formulates deployment plans for co-existing NFs on heterogeneous devices. Its modular NF library and user-friendly interface make the system accessible to non-experts. *(Published at NFV-SDN 2025; journal extension under review at IEEE/ACM ToN.)*

PROFESSIONAL EXPERIENCE

Shanghai Institute of Microsystem and Information Technology September 2021–September 2022
Chinese Academy of Sciences — Research Intern, Shanghai, China
Improved packet-forwarding throughput in intelligent edge gateway network drivers using a user-space data path and adaptive small-packet aggregation, and developed an edge-cloud collaborative system for real-time queries across multiple video streams. This work led to the SMC 2022, APNOMS 2022, and SMC 2023 papers above.

ACADEMIC ACHIEVEMENTS AND RECOGNITION

- Outstanding Student Award, East China Normal University November 2022
- Outstanding Student Award, East China Normal University November 2021
- Second Prize, 17th China Graduate Mathematical Contest in Modeling November 2020

TEACHING EXPERIENCE

University of Connecticut Spring 2027
Teaching Assistant, CSE 5300: Advanced Computer Networks

University of Connecticut Fall 2024, Spring 2025, Spring 2026
Teaching Assistant, CSE 2050: Data Structures and Object-Oriented Design

East China Normal University Fall 2022
Teaching Assistant, The Practice of Object-Oriented Programming (Java)

East China Normal University Fall 2021
Teaching Assistant, Advanced Mathematics